

SMART INDIA HACKATHON 2026



Problem Statement ID **SIH 26160**

Problem Statement Title **AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework**

Theme **Blockchain & Cybersecurity**

PS Category **Software**

Team ID **SIH 163**

Team Name **Highlanders**

One capture in — graded, cited evidence out

THE PROBLEM

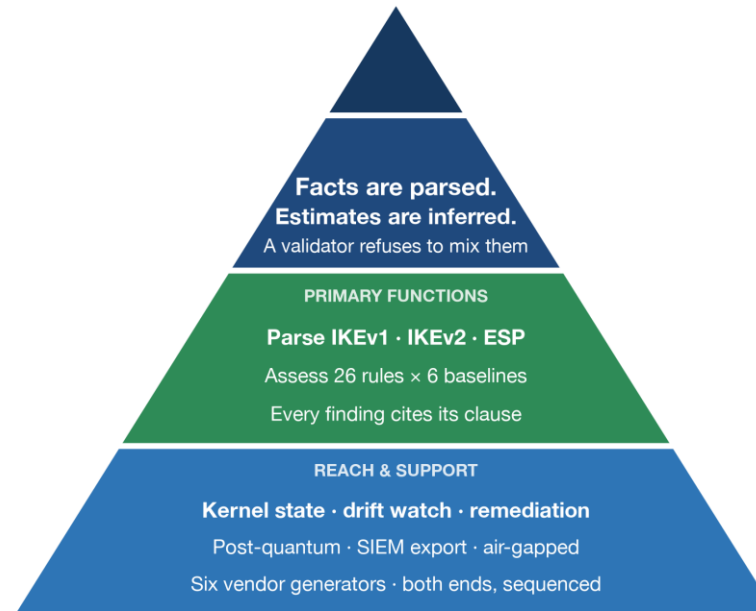
Wireshark decodes an IKE exchange. It has no opinion about it, so weak IPsec survives for years.

THE IDEA

26 deterministic rules judge what was negotiated, each citing the clause it applies.

WHAT IS NEW

Facts are parsed, estimates inferred — and a validator refuses to mix them.



Depth of capability, from one principle

GAP

Wireshark shows,
never judges

Rekey is encrypted —
tunnel goes dark

Payload is encrypted —
what does it carry?

A one-sided fix
is an outage

WHAT WE BUILT

26 cited rules,
clause by clause

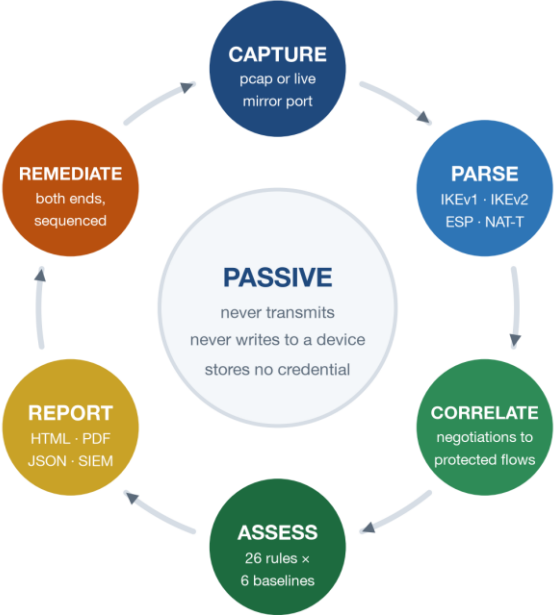
Kernel + daemon state
read and graded

Flow-shape inference,
calibrated, abstains

Both ends, sequenced,
zero packet loss

Two boundaries the design will not cross: it never writes to a network device, and stores no credential — an AST sweep over every module enforces both.

How it works, and what it is built from



Drop a capture here, or choose one.

Choose File 01-worst-ike...es-voip.pcap

This tool reads the capture and produces documents. It never writes to a network device and stores no credential. Reports are held in memory only.

Overview

Inventory

Findings

Exposure

Threat matrix

Post-quantum

Remediation

Estate overview

The tunnel examined is protected by cryptography that is no longer considered safe, and should be changed. The estate scores 0 out of 100 (grade F).

F
GRADE

0
SCORE / 100

1
TUNNELS

0
UNDOCUMENTED

3
CRITICAL

2
HIGH

3
MEDIUM

SECTION A — VERIFIED

8 finding(s) read directly from the negotiation. Facts; no confidence value applies.

SECTION B — INFERRED

0 finding(s) estimated from traffic patterns by a statistical model. Every one states a confidence and none is compliance evidence.

- No documented tunnel list was supplied, so this report cannot say whether any of these tunnels are expected to exist.
- 8 findings in Section A are read directly from the negotiation and can be relied on as evidence. 0 findings in Section B are estimated from traffic patterns and carry a stated confidence.

- 1

PARSE

scapy-free streaming reader
- 2

MODEL

pydantic v2 · strict schemas
- 3

LEARN

scikit-learn · SHAP
- 4

SERVE

FastAPI · Jinja2 · WeasyPrint
- 5

PROVE

pytest · mypy --strict · ruff
- 6

TESTBED

strongSwan · Docker · netem

Working prototype — the dashboard on a real estate capture

Three tunnels, two weak. Grade F, every finding cited.

IKEv1 · IKEv2 · ESP

26 rules × 6 baselines

6 vendor generators

NAT-T 4500 · both registries

FEASIBILITY AND VIABILITY

What makes it work, what could go wrong

FEASIBILITY

- Working prototype, v1.0.0 tagged
- Runs on a laptop: 0.68 GB container
- No agent, no credential, no outage

VIABILITY

- Reads captures operators already have
- Air-gapped install, offline bundle
- ITSAR + CERT-In: no rival ships these

CHALLENGES & RISKS

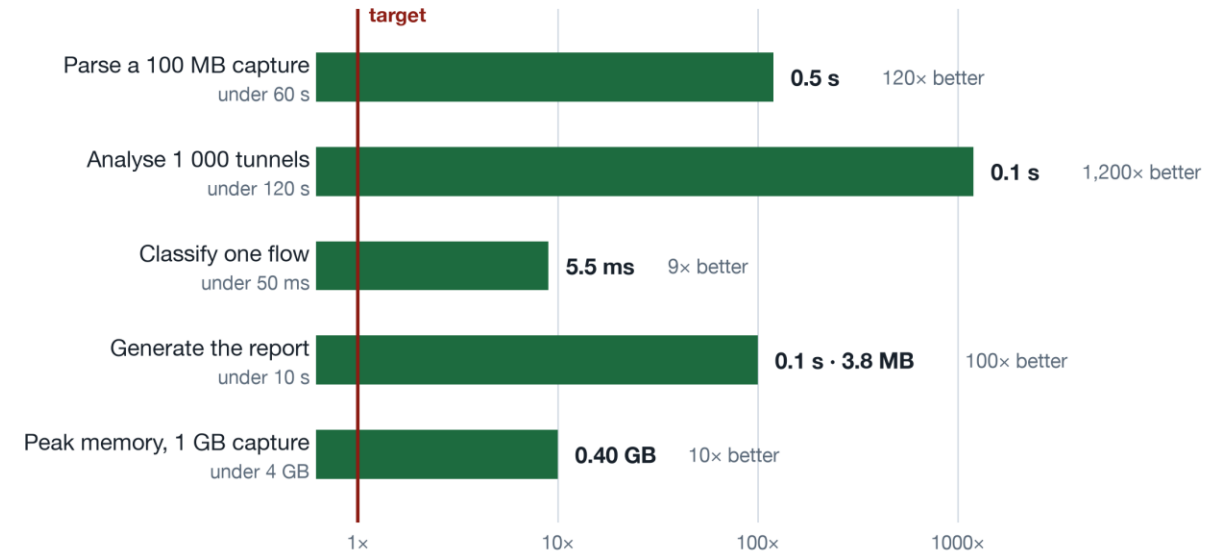
- Model trained on laboratory traffic
- 5 of 6 vendor configs not device-tested
- A capture can miss an encrypted rekey

STRATEGIES

- Publish held-out numbers, worst included
- Every generated file states its own status
- Read kernel + daemon state to close it

Every performance target beaten

Margin over the required limit, measured on the bundled captures



2 613

unit tests

306

integration, live pairs

93.8%

coverage

20/20

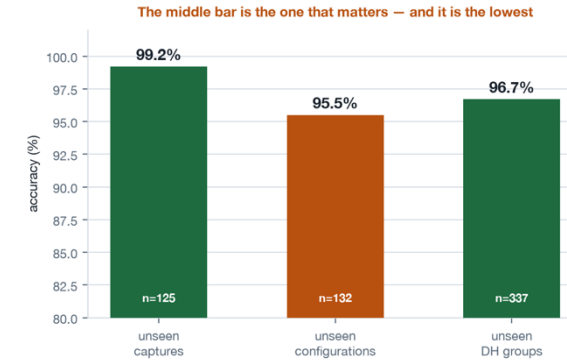
M11 acceptance

Deployable today

0.68 GB container · non-root · runs under --network none · offline bundle

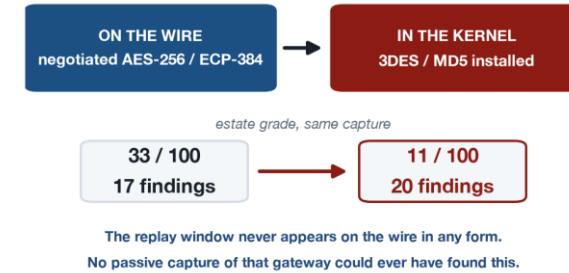
IMPACT AND BENEFITS

Who it helps, and what it changes



Honest about its limits

637 rows, 7 classes of laboratory traffic. The unflattering number is published, not the best one.



Endpoint visibility

Kernel state parsed and graded against the same rules — never fetched, never with a credential.

Social and economic benefit

Government, telecom and enterprise VPN estates get audit-grade evidence from a capture they already have — no agent, no credential, no outage, and an Indian baseline (ITSAR, CERT-In) that no comparable tool ships.

The standards it applies, and the data it learned from



Every one of the 26 rules cites the clause it applies — docs/RULES.md is generated from the registry

DATASETS — named in the problem statement

CIC-IDS2017 · CSE-CIC-IDS2018 · UNSW-NB15 · CTU-13 · CICIoT2023 · LANL · DARPA

Audited packet by packet, no sampling: none contains IKE, ESP or AH, so none carries a crypto label. Used as replayed inner payload — real traffic shape, our configuration supplies the labels.

DATASETS — generated for this project

strongSwan testbed corpus — 637 flow records · 7 classes · 36 configurations

Labels read back from what the tunnel actually negotiated. Every cipher meets at least two DH groups, so the classifier cannot learn the cipher instead of the traffic. Also used: ISCXVPN2016, MAWI, MITRE ATT&CK, NVD CVE.

Code: github.com/PrashamJ17/SIH_PS160 — v1.0.0 · M11 acceptance 20/20 · published JSON report schema 1.3

Evidence and demo: [Video Demo](#)